Tathagata Debnath

Ph.D. Candidate, Computer Science (NMSU, Las Cruces, USA)

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Skills

Bioinformatics (RNA-seq, Proteomics, Splicing) Expert

LLM Fine-Tuning Expert

PyTorch Expert

Multi-Omics Pipelines (rMATS, DESeq2, Trinity) Adv.

Multi-GPU Training Adv.

RAG & Vector DBs (FAISS) Adv.

Quantization / vLLM Adv.

Multi-Agent / LangGraph Adv.

Summary

PhD candidate at the intersection of **machine learning and biology**—building ML models for biological data and fine-tuning domain-specialized language models for the life sciences. **Biology:** computational analysis of alternative splicing (rMATS, *Scientific Reports* 2025), proteomics/phospho-proteomics pipelines, RNA-seq (Trinity, DESeq2), and circular-genome clustering algorithms (IEEE TPAMI, TCBB). **ML/DL:** end-to-end model training (PyTorch, QLoRA, DPO on multi-GPU), diffusion-adjacent generative methods, and RAG over scientific literature. Built **BioLite Suite** (biology-specialized SLMs on Llama 3.2) and **CodeQ** (MCTS+DPO, **91% fix rate**). IEEE TPAMI / TCBB / Nature author with **81+ citations** and 2 CRAN R packages.

Technical Stack

Bioinformatics: rMATS (splicing), DESeq2, Trinity (de novo assembly), STAR/HISAT2, MaxQuant / Proteome Discoverer, Gene Ontology / pathway enrichment, Bioconductor, samtools, bedtools

ML/DL: PyTorch, LoRA/QLoRA (PEFT), DPO/RLHF/SFT, Hugging Face (Transformers, TRL), Flash Attention 2, bitsandbytes (4-bit NF4), vLLM, scikit-learn, XGBoost, SHAP

LLM/Agents: LangGraph, LangChain, Claude/OpenAI APIs, RAG, FAISS, MCTS for Agentic Search

Infra: Multi-GPU Training (2×H100, NCCL), MIG GPU Partitioning (A100), Docker, Weights & Biases, LangSmith, FastAPI

Selected Projects

BioLite Suite: Domain-Specialized Biology SLMs **2026**
QLoRA, SFT, DPO, Llama 3.2, MIG A100

- Fine-tuned Llama 3.2 (1B/3B) with QLoRA on MIG-partitioned A100 (∼4.8 GB) for **biology narrative generation** from structured biological data.
- Released 2 Hugging Face models, 1 curated biology dataset, and full reproducible training pipeline.

NutriBot: Scientific Literature RAG System **2025 - 2026**
FAISS, BM25, RRF, FastAPI, Claude API

- Hybrid FAISS + BM25 + Reciprocal Rank Fusion retrieval over **nutrition and biomedical literature**.
- Cut per-query LLM cost ∼**50%** via caching, threshold gating, and token budgeting.

Federated Diabetes Predictor **2026**
Python, Flower, PyTorch, FedAvg

- Federated learning on **Diabetes Health Indicators** dataset, training across distributed clients without centralizing sensitive health data.
- Evaluated convergence and accuracy trade-offs vs. centralized baselines; designed for privacy-preserving clinical ML.

Classical ML (scikit-learn, XGBoost, CNN, RNN, GAN) Adv.

Python / R / C++ / SQL Expert

Education

Expected 2026

Ph.D., Computer Science

New Mexico State University

Advisor: Dr. Satyajayant Misra

Spring 2026

M.S., CS (AI/ML)

New Mexico State University

GPA: 4.00

2015 – 2017

M.Tech, CSE (Gold Medal)

Tripura University

GPA: 9.83/10

2011 – 2015

B.Tech, CSE

NIT Agartala

GPA: 7.62/10

Awards

· **M.Tech Gold Medal** (T.U.)
· **Biopattern Award**, (NMSU)
· **Ph.D. Tuition Scholarship**

Other Experience

Graduate Teaching Assistant
CS, NMSU
TA for Data Structures, Algorithms, AI, Machine Programming, C++, Java etc. courses.

CodeQ: Self-Improving Debugging Agent **2026**
PyTorch, DPO, MCTS, LoRA, TRL, 2×H100

- MCTS + DPO self-improvement on Qwen2.5-Coder-7B; **84% fix rate on DebugBench** (base 43.9%, MCTS-only 81.3%) after two DPO rounds.
- Built Fp32LogitsDPOTrainer to fix bf16 NaN overflow; dual-temperature prompt strategy (gen 0.8, critique 0.2).

Multi-Agent Code Generation Suite **2026**
LangGraph, Claude API, Docker, Streamlit, LLM-as-Judge

- Three progressively advanced multi-agent systems: baseline coder/tester loop, parallel DAG-based LangGraph workers with Docker-sandboxed pytest (**79/79 pass**), and self-evolving generator with LLM-as-Judge scoring.

Research Experience

Research Collaborator, Hansen Lab **2023 – 2025**
Dept. of Biology, NMSU

- Computational analysis of radiation-induced **alternative splicing** in *Aedes aegypti* via rMATS, identifying **197 differentially spliced genes** across Hippo and Notch signaling pathways (*Scientific Reports* 2025).
- Developed analysis workflows for multi-omics integration from RNA-seq data.

Research Collaborator, Castillo Lab **2021 – 2023**
Dept. of Biology, NMSU

- Built **proteomics / phospho-proteomics pipelines** (peptide quantification, PTM site mapping, differential abundance) and snail (*Physella acuta*) **RNA-seq pipelines** (alignment, Trinity de novo assembly, differential immune-response analysis).

Graduate Research Assistant **2020 – Present**
Dept. of Computer Science, NMSU

- Designed FOCC algorithm for optimal circular clustering on **round genomes** ($O(KN \log^2 N)$, IEEE/ACM TCBB); co-developed circular silhouette method (IEEE TPAMI); developing *Pohori* for alternative splicing detection.

Software

CRAN: OptCirClust · CircularSilhouette.

Selected Publications

- Bendzus-Mendoza H., Rodriguez A., **Debnath T.**, et al. "Radiation exposure induces genome-wide alternative splicing in *Aedes aegypti* mosquitoes." *Scientific Reports*, 2025.
- Chen Y., **Debnath T.**, Cai A., Song M. "Circular Silhouette and a Fast Algorithm." *IEEE TPAMI*, 45(11), 2023.
- **Debnath T.**, Song M. "Fast Optimal Circular Clustering on Round Genomes." *IEEE/ACM TCBB*, 18(6), 2021.
- Bhowmik M.K., **Debnath T.**, Bhattacharjee D., Dutta P. "EF-Index: Determining number of clusters (K) to estimate number of segments (S) in an image." *Image and Vision Computing*, 88:29–40, 2019.

Full list (81+ citations): Google Scholar *ORCID: 0000-0001-6445-275X*